UNIT 10 · TOPIC 6

Remote Sensing and Above-Ground Sensors

Syllabus topic: Remote sensing: aerial and satellite imagery; above-ground non-contact sensors

Remote sensing

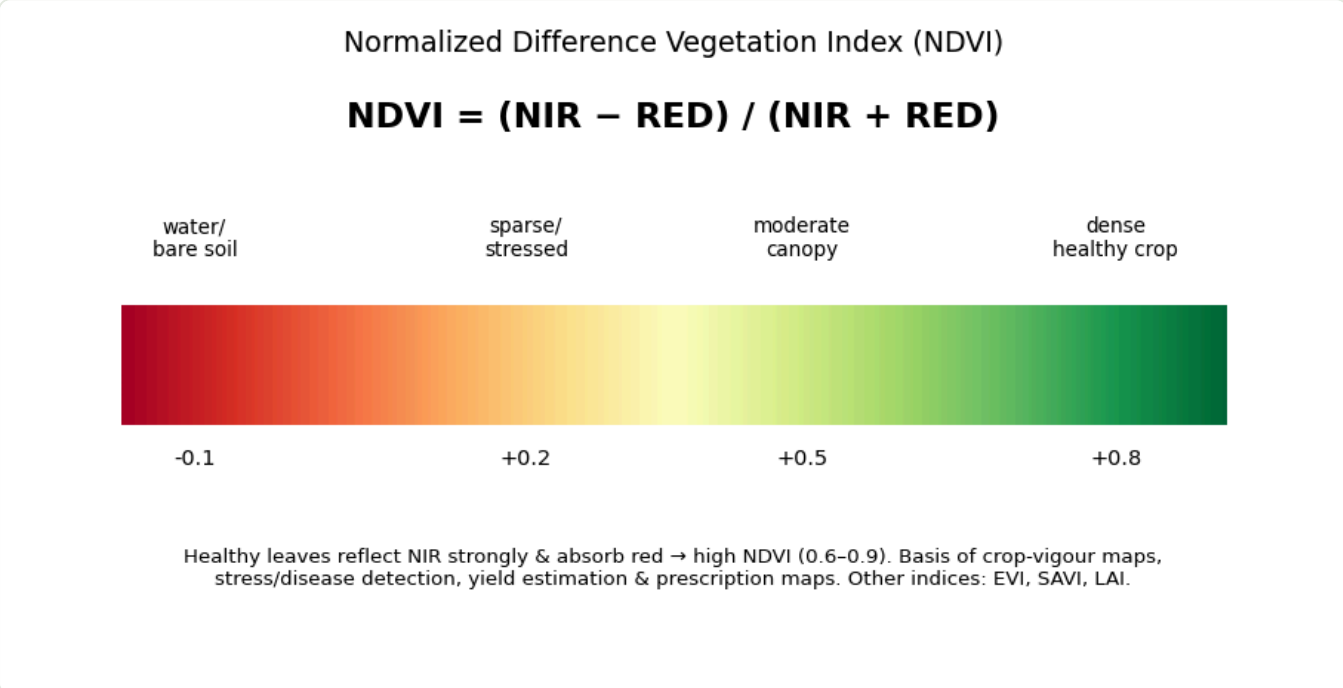
REMOTE SENSING acquires information about crops and soil WITHOUT contact, from AERIAL (aircraft, DRONE/UAV) or SATELLITE platforms sensing reflected or emitted radiation.

Sensor type	Energy source	Examples
PASSIVE	Reflected sunlight	Multispectral, hyperspectral, thermal cameras
ACTIVE	Emits its own energy	Radar, LiDAR

- **Platforms:** satellites (Landsat, Sentinel-2, India's IRS/Resourcesat), aircraft, and drones/UAVs — drones give the finest spatial detail on demand.
- Image resolutions and analysis: Topic 7.

Vegetation indices — NDVI

$$\text{NDVI} = (\text{NIR} - \text{RED}) \,/\, (\text{NIR} + \text{RED}) \text{ (range } -1 \text{ to } +1)$$



NDVI scale — water/bare soil to dense healthy crop

- Healthy vegetation REFLECTS near-infrared strongly and ABSORBS red → dense healthy crop gives high NDVI (~**0.6–0.9**); bare soil/water is near 0 or negative.
- NDVI maps crop vigour, stress, disease and biomass; it helps estimate yield and build prescription maps.
- Related indices: EVI, SAVI (soil-adjusted), LAI.

Above-ground (non-contact, proximal) sensors

- Mounted on the tractor/boom — e.g., optical crop-canopy "N-sensors" (GreenSeeker/Crop Circle type), multispectral/hyperspectral/thermal/LiDAR sensors.
- They read the crop in real time to drive SENSOR-BASED VRT (Topic 9).

Practice MCQs

Q44. Remote sensing acquires crop/soil information:

(a) By soil sampling
(b) Without contact, from aerial/satellite platforms
(c) By digging
(d) By weighing

Q49. A passive remote sensor uses:

(a) Its own radar pulse
(b) Reflected sunlight
(c) LiDAR
(d) Sound

Q50. LiDAR and radar are examples of ___ sensors:

(a) Passive
(b) Active
(c) Thermal-only
(d) Optical-only

Q51. The platform giving the highest spatial resolution on demand is the:

(a) Satellite
(b) High-altitude aircraft
(c) Drone/UAV
(d) Balloon

Q52. NDVI is calculated as:

(a) (RED-NIR)/(RED+NIR)
(b) (NIR-RED)/(NIR+RED)
(c) NIR×RED
(d) NIR/GREEN

Q53. A dense, healthy crop typically has NDVI of about:

(a) -0.5
(b) 0
(c) 0.6–0.9
(d) 5

Q54. NDVI is high for healthy vegetation because leaves:

(a) Absorb NIR
(b) Reflect NIR strongly and absorb red
(c) Reflect red
(d) Emit X-rays

Q55. Which is another vegetation index?

(a) SAVI
(b) BHN
(c) DOP
(d) GF

Q56. An above-ground 'N-sensor' on a boom is used for:

(a) Map-based VRT
(b) Sensor-based (real-time) VRT
(c) Yield weighing
(d) Steering

Answers

Q44 — (b) Without contact, from aerial/satellite platforms
Q49 — (b) Reflected sunlight
Q50 — (b) Active
Q51 — (c) Drone/UAV
Q52 — (b) (NIR-RED)/(NIR+RED) — *NDVI = (NIR - RED)/(NIR + RED).*
Q53 — (c) 0.6–0.9 — *Healthy crop 0.6–0.9.*
Q54 — (b) Reflect NIR strongly and absorb red — *Healthy crop reflects NIR and absorbs red.*
Q55 — (a) SAVI
Q56 — (b) Sensor-based (real-time) VRT

Sources: Srinivasan — Handbook of Precision Agriculture; ICAR/NIIT PA modules; Lillesand & Kiefer — Remote Sensing and Image Interpretation; ISO 11783 (ISOBUS); remote-sensing and NDVI references (as given in the source notes, July 2026); NPTEL precision-agriculture lectures.
